

Abhinav Goel

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Place of birth: Muzaffarnagar, Uttar Pradesh, India * *Date of birth:* 18-12-2004

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Education

B. Tech. in Electronics and Communication Engineering

PDPM IITDMJ

Bachelor's Degree Program

August 2024 - Present

Semester IV (CPI - 8.5)

Senior Secondary (Higher Secondary)

PR Public School, Muzaffarnagar

Central Board of Secondary Education (CBSE)

July 2022 - July 2024

Percentage: 93.6%

Secondary (High School)

Shardein School, Muzaffarnagar

Central Board of Secondary Education (CBSE)

April 2020 - July 2022

Percentage: 98.4%

Areas of interest

- Autonomous Mobile Robots (AMRs), SLAM and Navigation (ROS 2, Nav2)
- Unmanned Aerial Vehicles (UAVs) & Autonomous Aerial Flight Systems
- Robotic Manipulation, Visual Servoing & Kinematics
- Computer Vision, Spatial Geolocation & Multi-Spectral Perception
- Tactical Ground Control Stations (GCS) & Human-Machine Interfaces (Qt6 / C++)
- Embedded Systems, Flight Controllers & Discrete Power Electronics
- Subterranean, Field Exploration & Bio-Inspired Legged Robotics

Honors and key achievements

- **e-Yantra Robotics Competition (e-YRC 2025–26), IIT Bombay:** Top 10 National Finalist among 400+ participating teams across India in the *KrishiCobot* theme, designing an autonomous agricultural harvesting mobile manipulator with ROS 2, UR5 visual servoing, and precision vision pipelines. [[Certificate](#)]
- **DD Robocon 2025 (Doordarshan & IIT Delhi):** Achieved Stage-2 National Qualification representing Team Barbarick with an evaluation score of 86/100, designing competition-grade holonomic robot chassis and actuation mechanisms. [[Certificate](#)]
- **SAE Aerothon 2025 (SAE India):** Phase 2 National Qualification in the premier aerial robotics challenge, demonstrating an autonomous fixed-altitude surveillance quadcopter with ground target pixel-to-GPS conversion. [[Certificate](#)]
- **NIDAR 2026 (National Innovation Challenge for Drone Application and Research):** National Finalist, Aero Fabrication Club, developing an autonomous disaster surveillance UAV with 75+ acre automated polygon coverage and real-time survivor detection. [[Certificate](#)]

Autonomous Agricultural Harvesting Robot (KrishiCobot)

August 2025 - May 2026

e-Yantra Robotics Competition 2025–26, IIT Bombay

github.com/Humobot1812/Krishi_Cobot

- Architected and developed a full-stack autonomous harvesting robot integrating an AMR mobile base with a 6-DOF UR5 robotic arm, operating autonomously in complex orchard rows within Gazebo Ignition.
- Constructed a multi-sensor navigation pipeline fusing 2D planar LiDAR, ultrasonic range arrays, IMU, and wheel odometry inside ROS 2 Nav2, utilizing AMCL localization to achieve repeatable navigation in narrow crop corridors.
- Developed a two-tier perception pipeline integrating YOLOv8 for fruit classification and ArUco markers for 6-DOF pose estimation, projecting camera coordinates to 3D robot base frame with millimeter precision.
- Engineered Cartesian trajectory generation via MoveIt 2 alongside Image-Based Visual Servoing (IBVS) to continuously update end-effector paths in real time based on active camera tracking.
- Coordinated subsystems through a ROS 2 action server and behavior tree framework; achieved a Top 10 national rank out of 400+ teams across India.

Autonomous Warehouse Stock Auditing AMR & Active Vision System (WareOps)

December 2025 - August 2026

Robotics & Autonomous Systems Lab, PDPM IITDMJ

github.com/WareOps

- Engineered an autonomous mobile robot for automated warehouse stock auditing, featuring autonomous aisle navigation, active mast scanning, and automated digital ledger reporting.
- Configured the ROS 2 Nav2 stack with SLAM Toolbox for initial mapping and AMCL for runtime localization, tuning DWB local planner costmap critics for smooth traversal of narrow rack aisles.
- Designed and fabricated a 2-DOF pan-tilt motorized vision mast controlled by ROS 2 action servers to sweep multiple storage rack tiers during audit waypoints.
- Implemented a dual-engine QR reading pipeline executing OpenCV WeChat QR detector in parallel with PyZbar for high-accuracy barcode and QR decoding on damaged labels, logging results into an SQLite database.
- Maintained 1:1 hardware-simulation parity between Gazebo Ignition and physical hardware comprising Raspberry Pi 4, RPLidar A1M8, Arduino Nano via ros2_control, and ESP32-CAM Wi-Fi video streaming.

Autonomous 6-Wheeled Subterranean Exploration Rover (CAVE_DIVER)

September 2026 - Present

Electronics and Robotics Society, PDPM IITDMJ

github.com/Humobot1812/CAVE_DIVER_6_Wheeled_Rover

- Engineered an autonomous 6-wheeled robotic rover for exploration, 3D structural mapping, and environmental hazard assessment in GPS-denied cave and subterranean environments.
- Designed a rocker-bogie passive suspension mechanism modeled in Autodesk Fusion 360 and 3D-printed in PETG, enabling all 6 driven wheels to maintain continuous terrain contact across obstacles up to 1.5× wheel diameter.
- Developed a multi-spectral perception system featuring a 2-DOF gimbal carrying a low-light Raspberry Pi Camera v3 NoIR alongside a FLIR-compatible thermal sensor for human heat signature and hazard detection.
- Integrated an onboard 6-channel hazardous gas telemetry suite monitoring CO, H₂S, CH₄, LPG, and smoke concentrations at 1 Hz with automated threshold alarming and safety interlocks.
- Deployed SLAM Toolbox in lifelong mapping mode for 2D LiDAR-based mapping and pose estimation in GPS-denied subterranean tunnels, paired with a custom web-based Flask Ground Control Station.

Tactical Multi-UAV Ground Control Station (DroneGCS)

September 2026 - Present

Aero Fabrication Club, PDPM IITDMJ

github.com/Humobot1812/DroneGCS

- Developed an open-source tactical Ground Control Station in C++17 and Qt 6 / QML for simultaneous command, control, and telemetry tracking of multiple UAVs operating ArduPilot and PX4 autopilots.
- Engineered a real-time Heads-Up Display (HUD) parsing MAVLink v2 telemetry packets (HEARTBEAT, ATTITUDE, GPS_RAW_INT, SYS_STATUS, VFR_HUD) at 10 Hz, rendering pitch/roll artificial horizon, airspeed, altitude, and heading compass.
- Integrated a dynamic OpenStreetMap interactive tactical map via QtLocation plugin, supporting real-time multi-UAV GPS position tracking, multi-waypoint mission editing, and polygon geofence deployment.
- Implemented asynchronous C++ network backends for MAVLink over UDP/TCP and serial connections, handling automated parameter management (PARAM_REQUEST_LIST / PARAM_SET) and command protocols.
- Integrated GStreamer-accelerated H.264 video pipelines for ultra-low latency live onboard camera streaming with multi-drone telemetry overlay.

Beginner-Safe ROS 2 Drone Controller Stack (SafePilot)

September 2026 - Present

Open-Source UAV Development

github.com/Humobot1812/SafePilot

- Created an open-source ROS 2 package wrapping MAVSDK to implement software-enforced flight safety envelopes and autonomous flight assistance for research and beginner UAV pilots.
- Developed a joystick teleoperation node with dynamic velocity and attitude clamps that restrict maximum bank angles, descent rates, and horizontal accelerations to prevent vehicle loss-of-control.
- Implemented a 10 Hz 3D cylindrical geofencing monitor that applies progressive resistive velocity damping as the vehicle nears boundary walls and triggers autonomous Return-to-Launch (RTL) upon perimeter breach.
- Integrated autonomous assistance flight modes including altitude-hold with planar joystick nudging and automated landing sequences with precision descent profiling.

Autonomous Disaster Surveillance UAV & Victim Detection

June 2025 - January 2026

NIDAR 2026 National UAV Challenge, Aero Fabrication Club

github.com/Humobot1812/Autonomous_drone

- Developed an autonomous aerial surveillance system for large-area disaster relief and victim localization, capable of autonomously covering 75+ acres within 20 minutes at 25 m altitude.
- Built a computational geometry pipeline using Shapely and KML boundary parsing to generate optimized boustrophedon (zigzag) survey flight paths with parameterized lane overlap.
- Deployed a fine-tuned YOLOv8 human detection model on the companion computer processing downward camera feeds at 15 FPS, triggering autonomous loiter and target-centering routines upon detection.
- Automated the complete mission lifecycle through MAVSDK Python, handling automated pre-arm checks, guided takeoff, survey path traversal, interrupt-driven victim loiter, geofence compliance, and failsafe RTL.
- Conducted extensive Software-In-The-Loop (SITL) validation in ArduPilot SITL with simulated visual streams prior to field trials.

Autonomous Aerial Survey & Ground Target Geolocation Drone

January 2025 - June 2025

SAE Aerothon 2025, Aero Fabrication Club, PDPM IITDMJ

github.com/Humobot1812/Aerothon-25

- Engineered an autonomous UAV platform selected for national-level presentation at SAE Aerothon 2025, integrating a Pixhawk flight controller, Raspberry Pi companion computer, and ArduPilot autopilot.
- Derived and implemented a pinhole camera projection algorithm converting 2D pixel coordinates of detected targets into georeferenced latitude/longitude coordinates accurate to within ± 3 m from 30 m flight altitude.
- Trained and deployed YOLOv4-Tiny on the companion computer for real-time target panel classification and bounding-box estimation at 12 FPS.
- Developed a visual PID control loop computing lateral and longitudinal velocity commands to center detected targets beneath the vehicle during loiter maneuvers.

Differential Drive Mobile Robot — ROS 2 Navigation & SLAM

June 2025 - July 2025

Electronics and Robotics Society, PDPM

github.com/Humobot1812/Differential-Drive-ROS2-Navigation-Project

IIITDMJ

- Modeled and simulated a 4-wheeled differential-drive autonomous robot from scratch using URDF/Xacro, integrating collision models, wheel inertia matrices, LiDAR, and IMU sensor plugins in Gazebo.
- Configured SLAM Toolbox in asynchronous mode to generate 2D occupancy grid maps in simulation mazes, tuning scan-matching thresholds for sharp wall boundaries and minimal odometry drift.
- Implemented AMCL particle filtering for real-time localization on saved maps, tuning covariance parameters for rapid recovery during kidnapping and slippage conditions.
- Configured Nav2 with NavFn global planner and DWB local planner, optimizing obstacle avoidance critics for smooth trajectory generation through narrow doorway clearances.
- Integrated robot_localization Extended Kalman Filter (EKF) fusing wheel odometry and IMU data to ensure drift-free state estimation; authored a 12-part public tutorial series earning 7 GitHub stars.

Smart Multi-Protocol ESP32 Robotics Transmitter & Receiver System

April 2026 - July 2026

Electronics and Robotics Society, PDPM

github.com/Humobot1812/Smart-Multi-Protocol-ESP32-Transmitter

IIITDMJ

- Designed and built a modular dual-ESP32 smart remote control transmitter supporting four switchable wireless protocols: NRF24L01+ (2.4 GHz low latency), ESP-NOW (peer-to-peer 250 m), Wi-Fi UDP, and Bluetooth Classic.
- Developed firmware utilizing the U8g2 graphics library on a 128×64 ST7920 graphical LCD, rendering real-time RSSI signal strength, battery levels, channel bar graphs, and an interactive menu.
- Incorporated an MPU6050 6-axis IMU enabling tilt-based gesture control, mapping physical transmitter pitch and roll directly into motor velocity and steering commands.
- Built a companion modular receiver unit outputting multichannel servo PWM and serial telemetry back to the handheld unit.

Custom ESP32-S3 Quadcopter Drone with Discrete MOSFET ESCs

January 2026 - February 2026

Aero Fabrication Club, PDPM IIITDMJ

youtube.com/@The_IIITians_World

- Designed and assembled custom 3-phase Electronic Speed Controllers (ESCs) using discrete N-channel MOSFETs and gate drivers, eliminating commercial off-the-shelf ESCs.
- Implemented 6-step trapezoidal commutation firmware on an ESP32-S3 microcontroller, achieving smooth BLDC motor speed control with configurable acceleration ramps.
- Programmed a 500 Hz flight control PID loop reading angular rates and accelerations from an MPU6050 sensor to stabilize roll, pitch, and yaw axes in real time.
- Integrated a BMP280 barometric pressure sensor with accelerometer Z-axis data through a complementary filter for reliable relative altitude estimation and hover stabilization.

Modular 5-DOF Robotic Arm with Geometric Inverse Kinematics

January 2025 - March 2025

Electronics and Robotics Society, PDPM IIITDMJ

github.com/Humobot1812/Robotic_arm

- Designed and 3D-printed a 5-DOF articulated serial robotic arm utilizing high-torque MG996R and DS3225 metal-gear servo actuators.
- Established Denavit-Hartenberg (DH) kinematic frames and derived closed-form geometric inverse kinematics equations for real-time Cartesian point-to-point positioning on an Arduino microcontroller.
- Implemented cubic polynomial trajectory interpolation to ensure smooth acceleration and deceleration profiles, eliminating mechanical vibrations during rapid motion.
- Designed a compliant parallel-jaw gripper capable of securely grasping objects up to 60 mm in diameter.

Serpentine Snake Robot Prototype with Sinusoidal Locomotion Gaits

October 2024 - December 2024

Electronics and Robotics Society, PDPM IIITDMJ

github.com/Humobot1812/Snake_Robot

- Designed and constructed an 8-segment bio-inspired snake robot using modular 3D-printed body links interconnected with high-torque servo joints.
- Implemented Hirose serpenoid curve mathematical equations for gait generation, generating traveling sinusoidal waves with controllable amplitude, frequency, and phase offset.
- Evaluated multiple locomotion modes including lateral undulation for planar movement, sidewinding for loose terrain, and concertina locomotion for crawling through confined PVC pipes.

- Engineered an automated student check-in/check-out system deployed in the institute hostel, utilizing MFRC522 RFID reader hardware interfaced with ESP8266 Wi-Fi microcontrollers.
- Developed a Python Flask REST backend backed by SQLite to record real-time gate entry and exit events with cryptographic timestamp verification.
- Built a responsive administrative dashboard providing hostel authorities with real-time student presence counts, entry violation alerts, and downloadable attendance reports.

Certifications

- **e-Yantra Robotics Competition (e-YRC 2025–26) Finalist:** Awarded by IIT Bombay for securing Top 10 nationally among 400+ collegiate teams in the autonomous KrishiCobot agricultural robotics theme (August 2025 – May 2026).
- **DD Robocon 2025 Stage-2 Certificate:** Awarded by Doordarshan & IIT Delhi for qualifying Stage 2 with Team Barbarick, scoring 86/100 in technical design and mechanism evaluations (2025).
- **SAE Aerothon 2025 National Aerial Robotics Certificate:** Awarded by SAE India for qualifying Phase 2 in the national autonomous UAV surveillance challenge (June 2025).
- **NIDAR 2026 National UAV Challenge:** Certified participant in National Innovation Challenge for Drone Application and Research (January 2026).
- **Hands-on Introduction to Autonomous Systems with ROS 2 & Embedded Hardware:** PDPM IITDMJ Student Technical Workshop (2024).

Technical skills

Programming Languages	Python, C++, C, JavaScript, Bash
Tools and Frameworks	ROS2 Humble, Nav2, ros2_control, SLAM Toolbox, AMCL, Gazebo Ignition/Classic, RViz2, MoveIt 2, ArduPilot SITL, MAVSDK, DroneKit, Qt 6, QML, Flask, OpenCV, PlatformIO
Hardware and Systems	Raspberry Pi 4/5, NVIDIA Jetson Nano, Pixhawk (2.4.8, 6C), Arduino (Uno, Nano, Mega), ESP32, ESP32-S3, ESP32-CAM, STM32, RPLidar A1M8, Discrete MOSFET ESCs, MPU6050/9250 IMU, BMP280
Technologies	Autonomous Mobile Robots, Simultaneous Localization and Mapping (SLAM), Robot Operating System (ROS 2), Unmanned Aerial Vehicles (UAVs), Ground Control Stations (GCS), Computer Vision, Visual Servoing, Object Detection (YOLOv8), MAVLink Protocol, Embedded Systems, IoT, Cyber Physical Systems

Language proficiency

English, Hindi	Full Professional Proficiency
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Memberships

- **Electronics and Robotics Society of PDPM IITDMJ** (Coordinator, August 2024 – March 2026)
- **Aero Fabrication Club of PDPM IITDMJ** (Core Committee Member, August 2024 – Present)